databricks

Building Lakeflow Spark Declarative Pipelines

LECTURE

Deploying a Pipeline to Production

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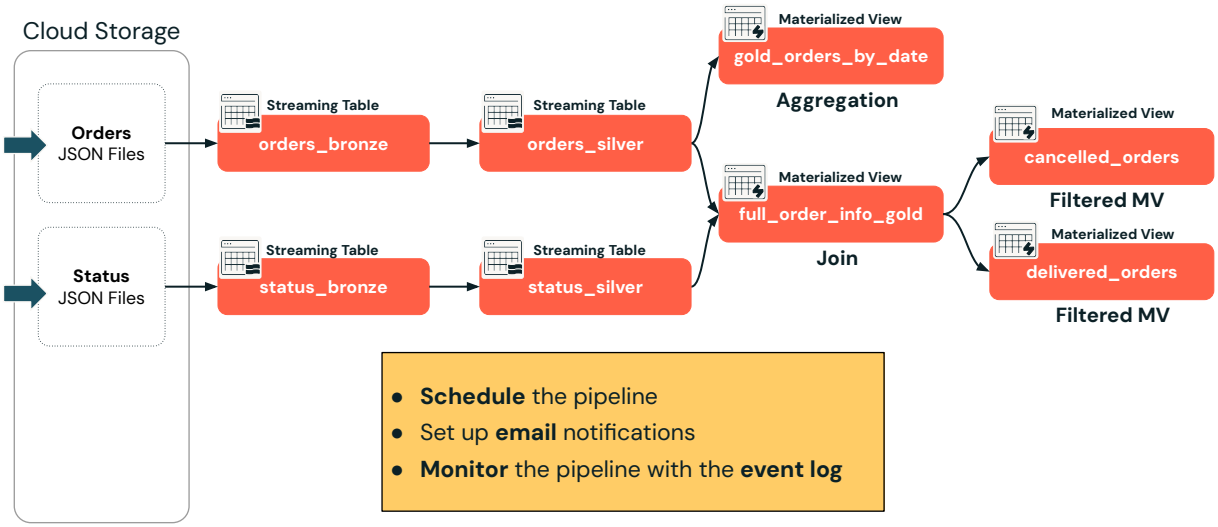
In this lecture, you will learn about common tasks involved in deploying a pipeline to production, focusing on operational best practices and requirements for reliable, production-grade data processing environments

Deploying a Pipeline to Production



- **Schedule** the pipeline
- Set up **email** notifications
- **Monitor** the pipeline with the **event log**

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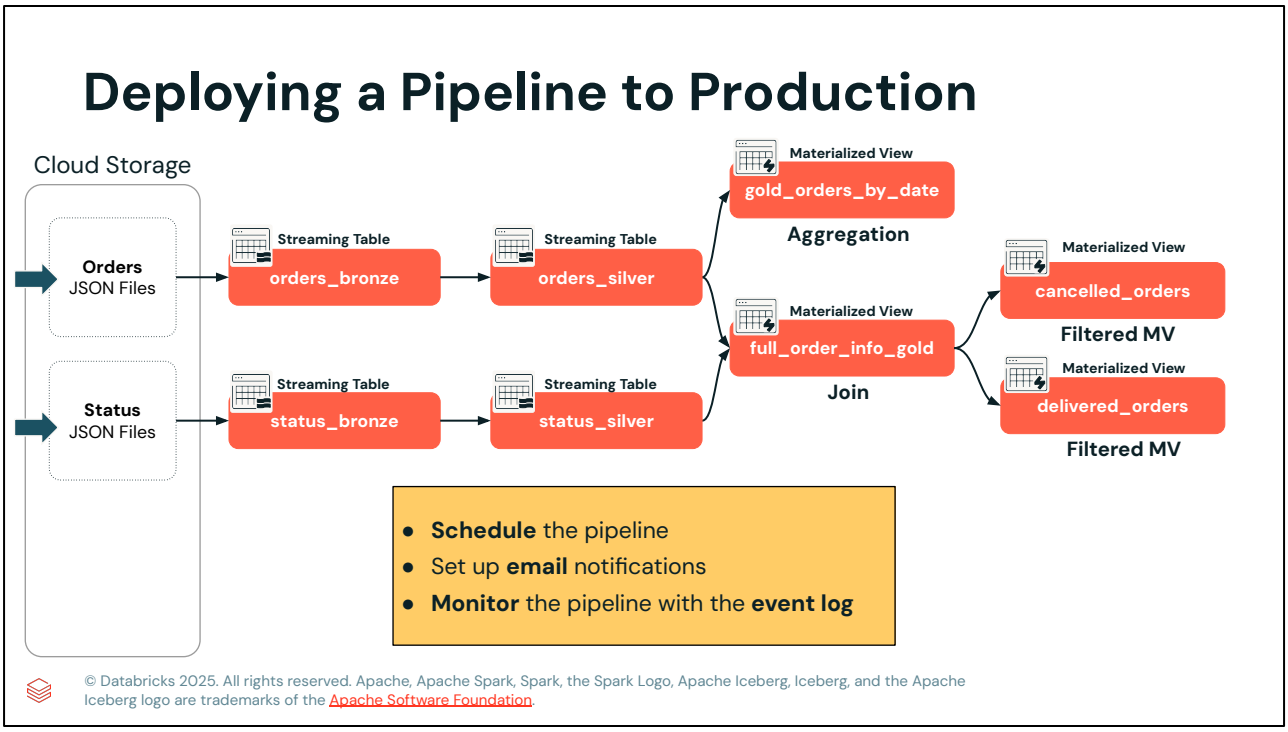
So far, we've completed the first flow of our pipeline:

Orders Flow: **orders JSON files → orders_bronze → orders_silver → gold_orders_by_date.** Now we're ready to add the second flow to the pipeline, the Status flow.

Here's what this new flow will do:

1. Ingest status JSON files to the **status_bronze** streaming table
2. Transform the bronze streaming table into the **status_silver** streaming table
3. Join **status_silver** with **orders_silver** to create a new materialized view: **full_order_info_gold**
4. From there, we'll build two additional materialized views:
   a. cancelled_orders
   b. Delivered_orders



This flow gives us rich insight into order lifecycle states, and demonstrates how we can join flows together and branch logic for more advanced analytics.

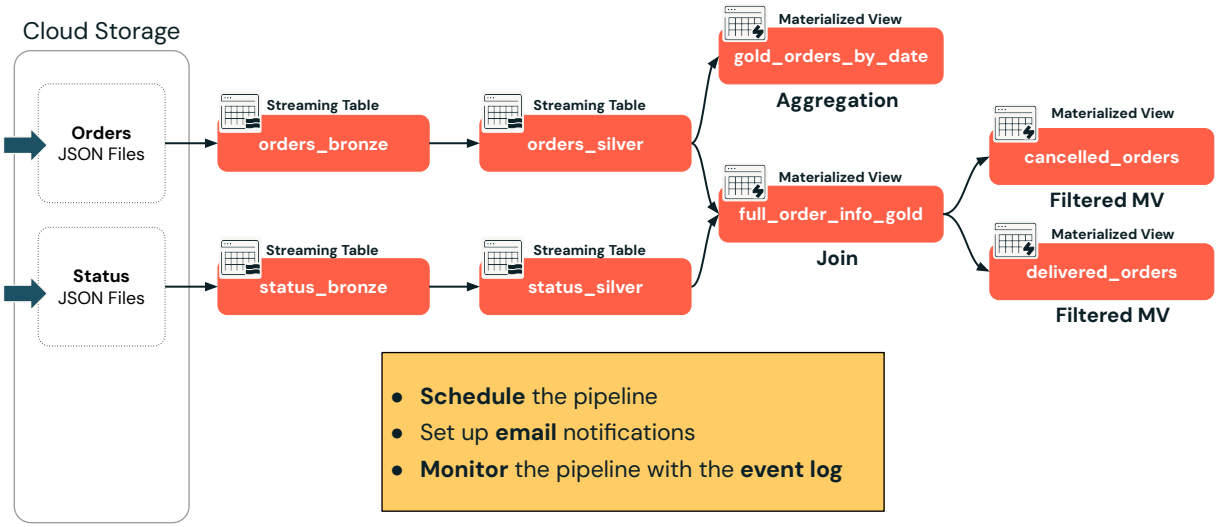
After we complete the Status Flow, our next step is to move the pipeline from development to production.

We'll walk through the key steps to operationalize the pipeline:

Schedule the Pipeline

- Define a schedule for automatic execution (e.g., every 15 minutes, hourly, daily, continuous).
- This ensures your data is refreshed regularly without manual triggers.

Deploying a Pipeline to Production



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Monitor the Pipeline with the Event Log

- Use the Event Log to view detailed runtime information.
- Track execution status, row counts, errors, warnings, and performance metrics.

With these steps, your pipeline will be fully production-ready, automated, monitored, and reliable.

Deploying a Pipeline to Production

Common Production Tasks

Scheduling

Email Notifications

Monitor pipelines

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When you're ready to move your Lakeflow Spark Declarative Pipeline into production, there are four key operational tasks to ensure it's reliable, automated, and monitored:

- Scheduling
- Email Notifications
- Monitor pipelines (event log)

Deploying a Pipeline to Production

Common Production Tasks

Scheduling

Email Notifications

Monitor pipelines

Triggered

- Can be triggered **manually** or run on a **schedule**
- **Refreshes** selected tables using data available at the start, then stops once the update is complete

Continuous

- **Continuously** processes new data to keep streaming tables and materialized views up to date in **near real time**
- Monitors dependencies and **updates only** when source data changes

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Once you've moved your pipeline into production mode, the next step is to schedule it to run automatically.

There are two primary scheduling options you can choose from, depending on your workload and latency requirements:

Triggered Mode

- Pipelines can be triggered manually or set to run on a recurring schedule.
- When triggered, the pipeline refreshes selected tables using the data available at the start of execution.
- Once updates are complete, the pipeline stops.
- This mode is ideal for batch processing or pipelines that don't need to run constantly.

Deploying a Pipeline to Production

Common Production Tasks

Scheduling

Email Notifications

Monitor pipelines

Triggered

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Continuous Mode

- Pipelines continuously process new data to keep streaming tables and materialized views up to date in near real-time.
- The system monitors dependencies and only updates when the source data changes.
- Best suited for streaming use cases where freshness and responsiveness are critical.

Deploying a Pipeline to Production

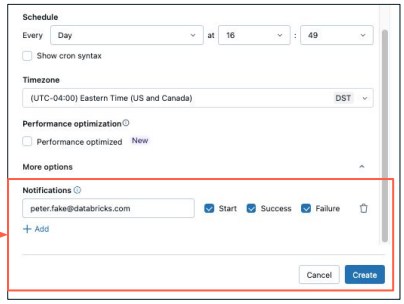
Common Production Tasks

Scheduling

Email Notifications

Monitor pipelines

You can configure **email notifications** when **scheduling** the pipeline



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As part of scheduling your pipeline, you can also configure email notifications to keep stakeholders informed of pipeline activity.

You have flexible options for when to trigger notifications. You can choose to receive an email:

- When the pipeline starts
- When the pipeline succeeds
- When the pipeline fails
- Or for any combination of the three events

This is a simple but powerful way to stay updated on your pipeline's health and status without having to constantly check the UI.

It's especially useful for production pipelines, so you're quickly alerted to issues or confirmed that everything ran smoothly.

Deploying a Pipeline to Production

Common Production Tasks

Scheduling

Email Notifications

Monitor pipelines

- The **pipeline event log** captures all key information about the pipeline:
  - Audit logs
  - Data Quality Checks
  - Pipeline Progress
  - Data Lineage

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Another essential task when running pipelines in production is monitoring their health and performance.

Spark Declarative Pipelines provide a pipeline event log that captures all critical information, including:

Audit Logs -Track who did what and when within the pipeline.

Data Quality Checks - Monitor the results of any expectations or constraints applied during processing.

Pipeline Progress - See the status and progress of each pipeline run.

Data Lineage - Understand how data flows and transforms through your pipeline steps.

This comprehensive event log helps you quickly diagnose issues, ensure data integrity, and maintain full visibility into your pipeline operations.

Deploying a Pipeline to Production

Querying the Declarative Pipeline Event Log

- **Publish the event log** as a **Delta table** using the **advanced settings**
  - Specify the table **location** (catalog, schema)
  - Define the **table** name

Query the pipeline event log table

```
SELECT * FROM <catalog>.<schema>.<event_log_table_name>
```

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The pipeline owner can publish the event log as a public Delta table for easier access and analysis.

This is done by toggling the “Publish event log to metastore” option found in the Advanced section of the pipeline configuration.

You can also optionally specify a custom table name, catalog, and schema for the event log table.

Once published, you can query the event log just like any other Delta table, enabling you to build custom reports, dashboards, or perform deeper analysis on pipeline activity and health.

Deploying a Pipeline to Production

Querying the Declarative Pipeline Event Log

- **Publish the event log** as a **Delta table** using the **advanced settings**
  - Specify the table **location** (catalog, schema)
  - Define the **table** name

- **(DEFAULT)** The Event Log is written as a **hidden Delta table**
  - Located in the pipeline **default catalog** and **schema**
  - View the query the event log documentation to access the hidden event log

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Before the ability to publish the event log to the metastore was introduced, the event log was saved as a hidden Delta table.

This hidden table is located in the pipeline's default catalog and schema.

If you want to query this hidden event log, you can find detailed instructions in the official documentation here: Query the event log

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Thank you for completing this lesson and continuing your journey to develop your skills with us.